Newman-Penrose Formalism Calculations

Stephani et al., Equation (21.24)

Metric

$$ds^2 = (\frac{a^2 r^2 \Delta \cos(\theta)^2 + \Delta}{r^2})\mathrm{d}r \otimes \mathrm{d}r + (\frac{a^2 r^2 \cos(\theta)^2 + 1}{r^2})\mathrm{d}\theta \otimes \mathrm{d}\theta + (\frac{a^2 r^2 \Delta \cos(\theta)^2 - (a^4 r^2 \cos(\theta)^2 + a^2)\sin(\theta)^2 + \Delta}{r^2 \Delta \sin(\theta)^2})\mathrm{d}\phi \otimes \mathrm{d}\phi$$
$$+ (-\frac{a^3 + a r^2 + (a^5 r^2 + a^3 r^4 - a^3 r^2 \Delta)\cos(\theta)^2 - a\Delta}{r^2 \Delta})\mathrm{d}\phi \otimes \mathrm{d}t + (-\frac{a^3 + a r^2 + (a^5 r^2 + a^3 r^4 - a^3 r^2 \Delta)\cos(\theta)^2 - a\Delta}{r^2 \Delta})\mathrm{d}t$$
$$\otimes \mathrm{d}\phi + (-\frac{a^4 + 2\,a^2 r^2 + r^4 + (a^6 r^2 + 2\,a^4 r^4 + a^2 r^6)\cos(\theta)^2 - (a^4 r^2 \Delta \cos(\theta)^2 + a^2 \Delta)\sin(\theta)^2}{r^2 \Delta})\mathrm{d}t \otimes \mathrm{d}t$$

Coordinates

$(x^\mu) = (r, \theta, \phi, t)$

Metric Functions

$\Delta = a^2 - e^2 + r^2 - 2\,m - r$

Null Tetrad

Covariant components

$$l = (1)\,dr + (\frac{a}{\Delta})\,d\phi + (\frac{a^2}{\Delta} + \frac{r^2}{\Delta})\,dt$$

$$n = (-\frac{1}{2}\,a^2 \Delta \cos(\theta)^2 - \frac{\Delta}{2\,r^2})\,dr + (\frac{1}{2}\,a^3 \cos(\theta)^2 + \frac{a}{2\,r^2})\,d\phi + (\frac{1}{2}\,a^4 \cos(\theta)^2 + \frac{1}{2}\,a^2 r^2 \cos(\theta)^2 + \frac{a^2}{2\,r^2} + \frac{1}{2})\,dt$$

$$m = (\frac{1}{2} i\,\sqrt{2} a \cos(\theta) + \frac{\sqrt{2}}{2\,r})\,d\theta + (-\frac{\sqrt{2} a \cos(\theta)}{2\,\sin(\theta)} + \frac{i\,\sqrt{2}}{2\,r \sin(\theta)})\,d\phi + (-\frac{1}{2}\,\sqrt{2} a^2 \cos(\theta)\sin(\theta) + \frac{i\,\sqrt{2} a \sin(\theta)}{2\,r})\,dt$$

$$\overline{m} = (-\frac{1}{2} i \sqrt{2} a \cos(\theta) + \frac{\sqrt{2}}{2\,r})\, d\theta + (-\frac{\sqrt{2} a \cos(\theta)}{2\,\sin(\theta)} - \frac{i\sqrt{2}}{2\,r\sin(\theta)})\, d\phi + (-\frac{1}{2}\sqrt{2} a^2 \cos(\theta)\sin(\theta) - \frac{i\sqrt{2} a \sin(\theta)}{2\,r})\, dt$$

Contravariant components

$$l = (\frac{r^2}{a^2 r^2 \Delta \cos(\theta)^2 + \Delta})\, dr + (\frac{a r^2 \sin(\theta)^2}{a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2})\, d\phi + (-\frac{r^2}{a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2})\, dt$$

$$n = (-\frac{1}{2})\, dr + (\frac{a \Delta \sin(\theta)^2}{2\,(a^2 \cos(\theta)^2 + r^2)})\, d\phi + (-\frac{\Delta}{2\,(a^2 \cos(\theta)^2 + r^2)})\, dt$$

$$\begin{aligned} m = {} & (\frac{i\,a r^2 \cos(\theta)}{\sqrt{2} a^2 r^2 \cos(\theta)^2 + \sqrt{2}} + \frac{r}{\sqrt{2} a^2 r^2 \cos(\theta)^2 + \sqrt{2}})\, d\theta + (-\frac{\sqrt{2} a^3 r^2 \cos(\theta)\sin(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)} - \frac{\sqrt{2} a r^4 \cos(\theta)\sin(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)} \\ & + \frac{i\sqrt{2} a^2 r \sin(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)} + \frac{i\sqrt{2} r^3 \sin(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)})\, d\phi \\ & + (\frac{\sqrt{2} a^2 r^2 \cos(\theta)\sin(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)} - \frac{i\sqrt{2} a r \sin(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)})\, dt \end{aligned}$$

$$\begin{aligned} \overline{m} = {} & (-\frac{i\,a r^2 \cos(\theta)}{\sqrt{2} a^2 r^2 \cos(\theta)^2 + \sqrt{2}} + \frac{r}{\sqrt{2} a^2 r^2 \cos(\theta)^2 + \sqrt{2}})\, d\theta + (-\frac{\sqrt{2} a^3 r^2 \cos(\theta)\sin(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)} - \frac{\sqrt{2} a r^4 \cos(\theta)\sin(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)} \\ & - \frac{i\sqrt{2} a^2 r \sin(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)} - \frac{i\sqrt{2} r^3 \sin(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)})\, d\phi \\ & + (\frac{\sqrt{2} a^2 r^2 \cos(\theta)\sin(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)} + \frac{i\sqrt{2} a r \sin(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)})\, dt \end{aligned}$$

Directional Derivatives

$$D = \frac{r^2}{a^2 r^2 \Delta \cos(\theta)^2 + \Delta}\, \partial_r + \frac{a r^2 \sin(\theta)^2}{a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2}\, \partial_\phi - \frac{r^2}{a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2}\, \partial_t$$

$$\Delta = -\frac{1}{2}\, \partial_r + \frac{a \Delta \sin(\theta)^2}{2\,(a^2 \cos(\theta)^2 + r^2)}\, \partial_\phi - \frac{\Delta}{2\,(a^2 \cos(\theta)^2 + r^2)}\, \partial_t$$

$$\delta = \frac{i\,ar^2\cos(\theta)}{\sqrt{2}a^2r^2\cos(\theta)^2+\sqrt{2}} + \frac{r}{\sqrt{2}a^2r^2\cos(\theta)^2+\sqrt{2}}\,\partial_\theta - \frac{\sqrt{2}a^3r^2\cos(\theta)\sin(\theta)}{2\left(a^4r^2\cos(\theta)^4+a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2\right)} - \frac{\sqrt{2}ar^4\cos(\theta)\sin(\theta)}{2\left(a^4r^2\cos(\theta)^4+a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2\right)}$$
$$+ \frac{i\sqrt{2}a^2r\sin(\theta)}{2\left(a^4r^2\cos(\theta)^4+a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2\right)} + \frac{i\sqrt{2}r^3\sin(\theta)}{2\left(a^4r^2\cos(\theta)^4+a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2\right)}\,\partial_\phi$$
$$+ \frac{\sqrt{2}a^2r^2\cos(\theta)\sin(\theta)}{2\left(a^4r^2\cos(\theta)^4+a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2\right)} - \frac{i\sqrt{2}ar\sin(\theta)}{2\left(a^4r^2\cos(\theta)^4+a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2\right)}\,\partial_t$$

$$\overline{\delta} = -\frac{i\,ar^2\cos(\theta)}{\sqrt{2}a^2r^2\cos(\theta)^2+\sqrt{2}} + \frac{r}{\sqrt{2}a^2r^2\cos(\theta)^2+\sqrt{2}}\,\partial_\theta - \frac{\sqrt{2}a^3r^2\cos(\theta)\sin(\theta)}{2\left(a^4r^2\cos(\theta)^4+a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2\right)} - \frac{\sqrt{2}ar^4\cos(\theta)\sin(\theta)}{2\left(a^4r^2\cos(\theta)^4+a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2\right)}$$
$$- \frac{i\sqrt{2}a^2r\sin(\theta)}{2\left(a^4r^2\cos(\theta)^4+a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2\right)} - \frac{i\sqrt{2}r^3\sin(\theta)}{2\left(a^4r^2\cos(\theta)^4+a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2\right)}\,\partial_\phi$$
$$+ \frac{\sqrt{2}a^2r^2\cos(\theta)\sin(\theta)}{2\left(a^4r^2\cos(\theta)^4+a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2\right)} + \frac{i\sqrt{2}ar\sin(\theta)}{2\left(a^4r^2\cos(\theta)^4+a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2\right)}\,\partial_t$$

Spin Coefficients

$$\rho = \frac{r}{a^4r^4\Delta\cos(\theta)^4+2\,a^2r^2\Delta\cos(\theta)^2+\Delta}$$

$$\sigma = \frac{i\,a^3r^4\cos(\theta)^3}{a^6r^4\cos(\theta)^6+a^4r^6\cos(\theta)^4+2\,a^4r^2\cos(\theta)^4+2\,a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2} + \frac{2\,a^2r^3\cos(\theta)^2}{a^6r^4\cos(\theta)^6+a^4r^6\cos(\theta)^4+2\,a^4r^2\cos(\theta)^4+2\,a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2}$$
$$- \frac{i\,ar^2\cos(\theta)}{a^6r^4\cos(\theta)^6+a^4r^6\cos(\theta)^4+2\,a^4r^2\cos(\theta)^4+2\,a^2r^4\cos(\theta)^2+a^2\cos(\theta)^2+r^2}$$

$$\kappa = -\frac{\sqrt{2}a^2r^5\cos(\theta)\sin(\theta)}{a^6r^4\Delta^2\cos(\theta)^6+a^4r^6\Delta^2\cos(\theta)^4+2\,a^4r^2\Delta^2\cos(\theta)^4+2\,a^2r^4\Delta^2\cos(\theta)^2+a^2\Delta^2\cos(\theta)^2+r^2\Delta^2}$$
$$+ \frac{i\sqrt{2}ar^4\sin(\theta)}{a^6r^4\Delta^2\cos(\theta)^6+a^4r^6\Delta^2\cos(\theta)^4+2\,a^4r^2\Delta^2\cos(\theta)^4+2\,a^2r^4\Delta^2\cos(\theta)^2+a^2\Delta^2\cos(\theta)^2+r^2\Delta^2}$$

$$\mu = \frac{1}{2\left(a^2r^3\cos(\theta)^2+r\right)}$$

$$\lambda = \frac{i\,a^3 r^2 \Delta \cos(\theta)^3}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)} - \frac{a^2 r \Delta \cos(\theta)^2}{a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2} - \frac{i\,a \Delta \cos(\theta)}{2\,(a^4 r^2 \cos(\theta)^4 + a^2 r^4 \cos(\theta)^2 + a^2 \cos(\theta)^2 + r^2)}$$

$$\nu = -\frac{\sqrt{2} a^2 r \cos(\theta) \sin(\theta)}{4\,(a^2 \cos(\theta)^2 + r^2)} - \frac{i\,\sqrt{2} a \sin(\theta)}{4\,(a^2 \cos(\theta)^2 + r^2)}$$

$$\begin{aligned}
\alpha = {} & \frac{3i\,a^5 r^4 \Delta \cos(\theta)^6}{2\,(\sqrt{2} a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2} a^4 r^6 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^4 r^2 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2} a^2 \Delta \cos(\theta)^2 + \sqrt{2} r^2 \Delta) \sin(\theta)} \\
& - \frac{2i\,a^5 r^4 \Delta \cos(\theta)^4}{(\sqrt{2} a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2} a^4 r^6 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^4 r^2 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2} a^2 \Delta \cos(\theta)^2 + \sqrt{2} r^2 \Delta) \sin(\theta)} \\
& + \frac{i\,a^3 r^6 \Delta \cos(\theta)^4}{2\,(\sqrt{2} a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2} a^4 r^6 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^4 r^2 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2} a^2 \Delta \cos(\theta)^2 + \sqrt{2} r^2 \Delta) \sin(\theta)} \\
& + \frac{a^4 r^5 \cos(\theta)^5}{2\,(\sqrt{2} a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2} a^4 r^6 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^4 r^2 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2} a^2 \Delta \cos(\theta)^2 + \sqrt{2} r^2 \Delta) \sin(\theta)} \\
& - \frac{a^4 r^3 \Delta \cos(\theta)^5}{(\sqrt{2} a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2} a^4 r^6 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^4 r^2 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2} a^2 \Delta \cos(\theta)^2 + \sqrt{2} r^2 \Delta) \sin(\theta)} \\
& - \frac{i\,a^3 r^6 \Delta \cos(\theta)^2}{(\sqrt{2} a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2} a^4 r^6 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^4 r^2 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2} a^2 \Delta \cos(\theta)^2 + \sqrt{2} r^2 \Delta) \sin(\theta)} \\
& - \frac{a^4 r^5 \cos(\theta)^3}{2\,(\sqrt{2} a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2} a^4 r^6 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^4 r^2 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2} a^2 \Delta \cos(\theta)^2 + \sqrt{2} r^2 \Delta) \sin(\theta)} \\
& + \frac{3\,a^4 r^3 \Delta \cos(\theta)^3}{2\,(\sqrt{2} a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2} a^4 r^6 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^4 r^2 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2} a^2 \Delta \cos(\theta)^2 + \sqrt{2} r^2 \Delta) \sin(\theta)} \\
& + \frac{i\,a^3 r^4 \cos(\theta)^4}{2\,(\sqrt{2} a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2} a^4 r^6 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^4 r^2 \Delta \cos(\theta)^4 + 2\,\sqrt{2} a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2} a^2 \Delta \cos(\theta)^2 + \sqrt{2} r^2 \Delta) \sin(\theta)} + \dots
\end{aligned}$$

$$
\begin{aligned}
\beta = {} & \frac{i\,a^5 r^4 \Delta \cos(\theta)^6}{2\left(\sqrt{2}a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2}a^4 r^6 \Delta \cos(\theta)^4 + 2\sqrt{2}a^4 r^2 \Delta \cos(\theta)^4 + 2\sqrt{2}a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2}a^2 \Delta \cos(\theta)^2 + \sqrt{2}r^2 \Delta\right)\sin(\theta)} \\
& - \frac{i\,a^5 r^4 \Delta \cos(\theta)^4}{\left(\sqrt{2}a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2}a^4 r^6 \Delta \cos(\theta)^4 + 2\sqrt{2}a^4 r^2 \Delta \cos(\theta)^4 + 2\sqrt{2}a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2}a^2 \Delta \cos(\theta)^2 + \sqrt{2}r^2 \Delta\right)\sin(\theta)} \\
& - \frac{i\,a^3 r^6 \Delta \cos(\theta)^4}{2\left(\sqrt{2}a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2}a^4 r^6 \Delta \cos(\theta)^4 + 2\sqrt{2}a^4 r^2 \Delta \cos(\theta)^4 + 2\sqrt{2}a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2}a^2 \Delta \cos(\theta)^2 + \sqrt{2}r^2 \Delta\right)\sin(\theta)} \\
& + \frac{a^4 r^5 \cos(\theta)^5}{2\left(\sqrt{2}a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2}a^4 r^6 \Delta \cos(\theta)^4 + 2\sqrt{2}a^4 r^2 \Delta \cos(\theta)^4 + 2\sqrt{2}a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2}a^2 \Delta \cos(\theta)^2 + \sqrt{2}r^2 \Delta\right)\sin(\theta)} \\
& - \frac{a^4 r^5 \cos(\theta)^3}{2\left(\sqrt{2}a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2}a^4 r^6 \Delta \cos(\theta)^4 + 2\sqrt{2}a^4 r^2 \Delta \cos(\theta)^4 + 2\sqrt{2}a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2}a^2 \Delta \cos(\theta)^2 + \sqrt{2}r^2 \Delta\right)\sin(\theta)} \\
& - \frac{a^4 r^3 \Delta \cos(\theta)^3}{2\left(\sqrt{2}a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2}a^4 r^6 \Delta \cos(\theta)^4 + 2\sqrt{2}a^4 r^2 \Delta \cos(\theta)^4 + 2\sqrt{2}a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2}a^2 \Delta \cos(\theta)^2 + \sqrt{2}r^2 \Delta\right)\sin(\theta)} \\
& - \frac{a^2 r^5 \Delta \cos(\theta)^3}{\left(\sqrt{2}a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2}a^4 r^6 \Delta \cos(\theta)^4 + 2\sqrt{2}a^4 r^2 \Delta \cos(\theta)^4 + 2\sqrt{2}a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2}a^2 \Delta \cos(\theta)^2 + \sqrt{2}r^2 \Delta\right)\sin(\theta)} \\
& - \frac{i\,a^3 r^4 \cos(\theta)^4}{2\left(\sqrt{2}a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2}a^4 r^6 \Delta \cos(\theta)^4 + 2\sqrt{2}a^4 r^2 \Delta \cos(\theta)^4 + 2\sqrt{2}a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2}a^2 \Delta \cos(\theta)^2 + \sqrt{2}r^2 \Delta\right)\sin(\theta)} \\
& + \frac{i\,a^3 r^2 \Delta \cos(\theta)^4}{\left(\sqrt{2}a^6 r^4 \Delta \cos(\theta)^6 + \sqrt{2}a^4 r^6 \Delta \cos(\theta)^4 + 2\sqrt{2}a^4 r^2 \Delta \cos(\theta)^4 + 2\sqrt{2}a^2 r^4 \Delta \cos(\theta)^2 + \sqrt{2}a^2 \Delta \cos(\theta)^2 + \sqrt{2}r^2 \Delta\right)\sin(\theta)} + \dots
\end{aligned}
$$

$$
\tau = \frac{i\,a^3 r^4 \cos(\theta)^2 \sin(\theta)}{\sqrt{2}a^4 r^4 \cos(\theta)^4 + 2\sqrt{2}a^2 r^2 \cos(\theta)^2 + \sqrt{2}} + \frac{a^2 r^3 \cos(\theta)\sin(\theta)}{\sqrt{2}a^4 r^4 \cos(\theta)^4 + 2\sqrt{2}a^2 r^2 \cos(\theta)^2 + \sqrt{2}}
$$

$$
\pi = \frac{i\,a^3 r^4 \cos(\theta)^2 \sin(\theta)}{\sqrt{2}a^4 r^4 \cos(\theta)^4 + 2\sqrt{2}a^2 r^2 \cos(\theta)^2 + \sqrt{2}} - \frac{a^2 r^3 \cos(\theta)\sin(\theta)}{\sqrt{2}a^4 r^4 \cos(\theta)^4 + 2\sqrt{2}a^2 r^2 \cos(\theta)^2 + \sqrt{2}}
$$

$$
\begin{aligned}
\epsilon = &-\frac{a^4 r^4 \cos(\theta)^4 \frac{\partial}{\partial r}\Delta}{2\,(a^6 r^4 \Delta^2 \cos(\theta)^6 + a^4 r^6 \Delta^2 \cos(\theta)^4 + 2\,a^4 r^2 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^4 \Delta^2 \cos(\theta)^2 + a^2 \Delta^2 \cos(\theta)^2 + r^2 \Delta^2)} \\
&+ \frac{i\,a^3 r^4 \Delta^2 \cos(\theta)^3}{2\,(a^6 r^4 \Delta^2 \cos(\theta)^6 + a^4 r^6 \Delta^2 \cos(\theta)^4 + 2\,a^4 r^2 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^4 \Delta^2 \cos(\theta)^2 + a^2 \Delta^2 \cos(\theta)^2 + r^2 \Delta^2)} \\
&- \frac{a^2 r^6 \cos(\theta)^2 \frac{\partial}{\partial r}\Delta}{2\,(a^6 r^4 \Delta^2 \cos(\theta)^6 + a^4 r^6 \Delta^2 \cos(\theta)^4 + 2\,a^4 r^2 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^4 \Delta^2 \cos(\theta)^2 + a^2 \Delta^2 \cos(\theta)^2 + r^2 \Delta^2)} \\
&+ \frac{a^2 r^5 \Delta \cos(\theta)^2}{a^6 r^4 \Delta^2 \cos(\theta)^6 + a^4 r^6 \Delta^2 \cos(\theta)^4 + 2\,a^4 r^2 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^4 \Delta^2 \cos(\theta)^2 + a^2 \Delta^2 \cos(\theta)^2 + r^2 \Delta^2} \\
&+ \frac{i\,a^3 r^2 \Delta \cos(\theta)^3}{2\,(a^6 r^4 \Delta^2 \cos(\theta)^6 + a^4 r^6 \Delta^2 \cos(\theta)^4 + 2\,a^4 r^2 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^4 \Delta^2 \cos(\theta)^2 + a^2 \Delta^2 \cos(\theta)^2 + r^2 \Delta^2)} \\
&+ \frac{i\,a r^4 \Delta \cos(\theta)}{2\,(a^6 r^4 \Delta^2 \cos(\theta)^6 + a^4 r^6 \Delta^2 \cos(\theta)^4 + 2\,a^4 r^2 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^4 \Delta^2 \cos(\theta)^2 + a^2 \Delta^2 \cos(\theta)^2 + r^2 \Delta^2)} \\
&- \frac{a^2 r^2 \cos(\theta)^2 \frac{\partial}{\partial r}\Delta}{2\,(a^6 r^4 \Delta^2 \cos(\theta)^6 + a^4 r^6 \Delta^2 \cos(\theta)^4 + 2\,a^4 r^2 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^4 \Delta^2 \cos(\theta)^2 + a^2 \Delta^2 \cos(\theta)^2 + r^2 \Delta^2)} \\
&+ \frac{i\,a r^2 \Delta^2 \cos(\theta)}{2\,(a^6 r^4 \Delta^2 \cos(\theta)^6 + a^4 r^6 \Delta^2 \cos(\theta)^4 + 2\,a^4 r^2 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^4 \Delta^2 \cos(\theta)^2 + a^2 \Delta^2 \cos(\theta)^2 + r^2 \Delta^2)} \\
&- \frac{r^4 \frac{\partial}{\partial r}\Delta}{2\,(a^6 r^4 \Delta^2 \cos(\theta)^6 + a^4 r^6 \Delta^2 \cos(\theta)^4 + 2\,a^4 r^2 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^4 \Delta^2 \cos(\theta)^2 + a^2 \Delta^2 \cos(\theta)^2 + r^2 \Delta^2)} \\
&+ \frac{r^3 \Delta}{a^6 r^4 \Delta^2 \cos(\theta)^6 + a^4 r^6 \Delta^2 \cos(\theta)^4 + 2\,a^4 r^2 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^4 \Delta^2 \cos(\theta)^2 + a^2 \Delta^2 \cos(\theta)^2 + r^2 \Delta^2}
\end{aligned}
$$

$$
\begin{aligned}
\gamma = &\frac{i\,a^3 r^3 \Delta \cos(\theta)^3}{4\,(a^4 r^3 \cos(\theta)^4 + a^2 r^5 \cos(\theta)^2 + a^2 r \cos(\theta)^2 + r^3)} + \frac{a^2 r^4 \cos(\theta)^2}{2\,(a^4 r^3 \cos(\theta)^4 + a^2 r^5 \cos(\theta)^2 + a^2 r \cos(\theta)^2 + r^3)} - \frac{i\,a^3 r \cos(\theta)^3}{4\,(a^4 r^3 \cos(\theta)^4 + a^2 r^5 \cos(\theta)^2 + a^2 r \cos(\theta)^2 + r^3)} \\
&- \frac{i\,a r^3 \cos(\theta)}{4\,(a^4 r^3 \cos(\theta)^4 + a^2 r^5 \cos(\theta)^2 + a^2 r \cos(\theta)^2 + r^3)} + \frac{i\,a r \Delta \cos(\theta)}{4\,(a^4 r^3 \cos(\theta)^4 + a^2 r^5 \cos(\theta)^2 + a^2 r \cos(\theta)^2 + r^3)} - \frac{a^2 \cos(\theta)^2}{2\,(a^4 r^3 \cos(\theta)^4 + a^2 r^5 \cos(\theta)^2 + a^2 r \cos(\theta)^2 + r^3)}
\end{aligned}
$$

Ricci Tensor Components

$$
\Phi_{00} = -\frac{a^6 r^8 \Delta^3 \cos(\theta)^6}{D_2} - \frac{2\,a^6 r^{10} \cos(\theta)^6}{D_2} + \frac{2\,a^6 r^{10} \cos(\theta)^4}{D_2} - \frac{2\,a^4 r^6 \Delta^3 \cos(\theta)^4}{D_2} - \dots
$$

$$
\begin{aligned}
D_2 = &\,a^{12} r^8 \Delta^3 \cos(\theta)^{12} + 2\,a^{10} r^{10} \Delta^3 \cos(\theta)^{10} + a^8 r^{12} \Delta^3 \cos(\theta)^8 + 4\,a^{10} r^6 \Delta^3 \cos(\theta)^{10} + 8\,a^8 r^8 \Delta^3 \cos(\theta)^8 + 4\,a^6 r^{10} \Delta^3 \cos(\theta)^6 + 6\,a^8 r^4 \Delta^3 \cos(\theta)^8 \\
&+ 12\,a^6 r^6 \Delta^3 \cos(\theta)^6 + 6\,a^4 r^8 \Delta^3 \cos(\theta)^4 + 4\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 8\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 4\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3
\end{aligned}
$$

$$\Phi_{01} = -\frac{3\,a^8 r^8 \Delta^3 \cos(\theta)^9}{2\sqrt{2}\,D_2 \sin(\theta)} - \frac{a^8 r^9 \cos(\theta)^9 \frac{\partial}{\partial r}\Delta}{\sqrt{2}\,D_2 \sin(\theta)} + \frac{5\,a^8 r^8 \Delta^3 \cos(\theta)^7}{2\sqrt{2}\,D_2 \sin(\theta)} - \dots$$

$$\Phi_{02} = \frac{a^{10} r^8 \Delta^2 \cos(\theta)^{10}}{D_3} - \frac{2\,a^{10} r^8 \Delta^2 \cos(\theta)^8}{D_3} + \frac{a^8 r^{10} \Delta^2 \cos(\theta)^8}{D_3} - \frac{a^8 r^8 \Delta^3 \cos(\theta)^8}{D_3} - \dots$$

$$\begin{aligned} D_3 = {} & a^{12} r^8 \Delta^2 \cos(\theta)^{12} + 2\,a^{10} r^{10} \Delta^2 \cos(\theta)^{10} + a^8 r^{12} \Delta^2 \cos(\theta)^8 + 4\,a^{10} r^6 \Delta^2 \cos(\theta)^{10} + 8\,a^8 r^8 \Delta^2 \cos(\theta)^8 + 4\,a^6 r^{10} \Delta^2 \cos(\theta)^6 + 6\,a^8 r^4 \Delta^2 \cos(\theta)^8 \\ & + 12\,a^6 r^6 \Delta^2 \cos(\theta)^6 + 6\,a^4 r^8 \Delta^2 \cos(\theta)^4 + 4\,a^6 r^2 \Delta^2 \cos(\theta)^6 + 8\,a^4 r^4 \Delta^2 \cos(\theta)^4 + 4\,a^2 r^6 \Delta^2 \cos(\theta)^2 + a^4 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^2 \cos(\theta)^2 + r^4 \Delta^2 \end{aligned}$$

$$\Phi_{10} = -\frac{3\,a^8 r^8 \Delta^3 \cos(\theta)^9}{2\sqrt{2}\,D_2 \sin(\theta)} - \frac{a^8 r^9 \cos(\theta)^9 \frac{\partial}{\partial r}\Delta}{\sqrt{2}\,D_2 \sin(\theta)} + \frac{5\,a^8 r^8 \Delta^3 \cos(\theta)^7}{2\sqrt{2}\,D_2 \sin(\theta)} - \dots$$

$$\Phi_{11} = -\frac{a^8 r^6 \Delta^3 \cos(\theta)^{10}}{4\,D_1 \sin(\theta)^2} - \frac{a^8 r^6 \cos(\theta)^{10} \frac{\partial}{\partial r}\Delta^2}{4\,D_1 \sin(\theta)^2} + \frac{a^8 r^6 \Delta \cos(\theta)^{10} \frac{\partial^2}{(\partial r)^2}\Delta}{8\,D_1 \sin(\theta)^2} + \frac{a^8 r^6 \Delta^3 \cos(\theta)^8}{4\,D_1 \sin(\theta)^2} + \frac{a^6 r^6 \Delta^4 \cos(\theta)^8}{4\,D_1 \sin(\theta)^2} + \dots$$

$$\begin{aligned} D_1 = {} & a^{10} r^6 \Delta^3 \cos(\theta)^{10} + 2\,a^8 r^8 \Delta^3 \cos(\theta)^8 + a^6 r^{10} \Delta^3 \cos(\theta)^6 + 3\,a^8 r^4 \Delta^3 \cos(\theta)^8 + 6\,a^6 r^6 \Delta^3 \cos(\theta)^6 + 3\,a^4 r^8 \Delta^3 \cos(\theta)^4 \\ & + 3\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 6\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 3\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3 \end{aligned}$$

$$\Phi_{12} = -\frac{3\,a^8 r^6 \Delta^3 \cos(\theta)^9}{4\sqrt{2}\,D_4 \sin(\theta)} - \frac{a^8 r^7 \cos(\theta)^9 \frac{\partial}{\partial r}\Delta}{2\sqrt{2}\,D_4 \sin(\theta)} + \frac{5\,a^8 r^6 \Delta^3 \cos(\theta)^7}{4\sqrt{2}\,D_4 \sin(\theta)} - \frac{a^6 r^8 \Delta^3 \cos(\theta)^7}{4\sqrt{2}\,D_4 \sin(\theta)} - \dots$$

$$\begin{aligned} D_4 = {} & a^{10} r^6 \Delta^2 \cos(\theta)^{10} + 2\,a^8 r^8 \Delta^2 \cos(\theta)^8 + a^6 r^{10} \Delta^2 \cos(\theta)^6 + 3\,a^8 r^4 \Delta^2 \cos(\theta)^8 + 6\,a^6 r^6 \Delta^2 \cos(\theta)^6 + 3\,a^4 r^8 \Delta^2 \cos(\theta)^4 \\ & + 3\,a^6 r^2 \Delta^2 \cos(\theta)^6 + 6\,a^4 r^4 \Delta^2 \cos(\theta)^4 + 3\,a^2 r^6 \Delta^2 \cos(\theta)^2 + a^4 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^2 \cos(\theta)^2 + r^4 \Delta^2 \end{aligned}$$

$$\Phi_{20} = \frac{a^{10} r^8 \Delta^2 \cos(\theta)^{10}}{D_3} - \frac{2\,a^{10} r^8 \Delta^2 \cos(\theta)^8}{D_3} + \frac{a^8 r^{10} \Delta^2 \cos(\theta)^8}{D_3} - \frac{a^8 r^8 \Delta^3 \cos(\theta)^8}{D_3} + \dots$$

$$\Phi_{21} = -\frac{3\,a^8 r^6 \Delta^3 \cos(\theta)^9}{4\sqrt{2}\,D_4 \sin(\theta)} - \frac{a^8 r^7 \cos(\theta)^9 \frac{\partial}{\partial r}\Delta}{2\sqrt{2}\,D_4 \sin(\theta)} + \frac{5\,a^8 r^6 \Delta^3 \cos(\theta)^7}{4\sqrt{2}\,D_4 \sin(\theta)} - \frac{a^6 r^8 \Delta^3 \cos(\theta)^7}{4\sqrt{2}\,D_4 \sin(\theta)} + \dots$$

$$
\begin{aligned}
\Phi_{22} = & \frac{a^6 r^4 \Delta^3 \cos(\theta)^8}{4\,(a^8 r^4 \Delta \cos(\theta)^8 + 2\,a^6 r^6 \Delta \cos(\theta)^6 + a^4 r^8 \Delta \cos(\theta)^4 + 2\,a^6 r^2 \Delta \cos(\theta)^6 + 4\,a^4 r^4 \Delta \cos(\theta)^4 + 2\,a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)\sin(\theta)^2} \\
& + \frac{a^6 r^6 \cos(\theta)^8}{2\,(a^8 r^4 \Delta \cos(\theta)^8 + 2\,a^6 r^6 \Delta \cos(\theta)^6 + a^4 r^8 \Delta \cos(\theta)^4 + 2\,a^6 r^2 \Delta \cos(\theta)^6 + 4\,a^4 r^4 \Delta \cos(\theta)^4 + 2\,a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)\sin(\theta)^2} \\
& - \frac{a^6 r^4 \Delta^3 \cos(\theta)^6}{4\,(a^8 r^4 \Delta \cos(\theta)^8 + 2\,a^6 r^6 \Delta \cos(\theta)^6 + a^4 r^8 \Delta \cos(\theta)^4 + 2\,a^6 r^2 \Delta \cos(\theta)^6 + 4\,a^4 r^4 \Delta \cos(\theta)^4 + 2\,a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)\sin(\theta)^2} \\
& - \frac{a^6 r^6 \cos(\theta)^6}{(a^8 r^4 \Delta \cos(\theta)^8 + 2\,a^6 r^6 \Delta \cos(\theta)^6 + a^4 r^8 \Delta \cos(\theta)^4 + 2\,a^6 r^2 \Delta \cos(\theta)^6 + 4\,a^4 r^4 \Delta \cos(\theta)^4 + 2\,a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)\sin(\theta)^2} \\
& + \frac{a^6 r^6 \cos(\theta)^4}{2\,(a^8 r^4 \Delta \cos(\theta)^8 + 2\,a^6 r^6 \Delta \cos(\theta)^6 + a^4 r^8 \Delta \cos(\theta)^4 + 2\,a^6 r^2 \Delta \cos(\theta)^6 + 4\,a^4 r^4 \Delta \cos(\theta)^4 + 2\,a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)\sin(\theta)^2} \\
& + \frac{a^4 r^2 \Delta^3 \cos(\theta)^6}{2\,(a^8 r^4 \Delta \cos(\theta)^8 + 2\,a^6 r^6 \Delta \cos(\theta)^6 + a^4 r^8 \Delta \cos(\theta)^4 + 2\,a^6 r^2 \Delta \cos(\theta)^6 + 4\,a^4 r^4 \Delta \cos(\theta)^4 + 2\,a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)\sin(\theta)^2} \\
& + \frac{3\,a^6 \Delta \cos(\theta)^8}{4\,(a^8 r^4 \Delta \cos(\theta)^8 + 2\,a^6 r^6 \Delta \cos(\theta)^6 + a^4 r^8 \Delta \cos(\theta)^4 + 2\,a^6 r^2 \Delta \cos(\theta)^6 + 4\,a^4 r^4 \Delta \cos(\theta)^4 + 2\,a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)\sin(\theta)^2} + \dots
\end{aligned}
$$

$$
\Lambda = -\frac{7\,a^8 r^6 \Delta^3 \cos(\theta)^{10}}{12\,D_1 \sin(\theta)^2} + \frac{a^8 r^6 \cos(\theta)^{10} \frac{\partial}{\partial r}\Delta^2}{12\,D_1 \sin(\theta)^2} - \frac{a^8 r^6 \Delta \cos(\theta)^{10} \frac{\partial^2}{(\partial r)^2}\Delta}{24\,D_1 \sin(\theta)^2} + \frac{13\,a^8 r^6 \Delta^3 \cos(\theta)^8}{12\,D_1 \sin(\theta)^2} - \frac{2\,a^6 r^8 \Delta^3 \cos(\theta)^8}{3\,D_1 \sin(\theta)^2} - \dots
$$

Weyl Tensor Components

$$
\begin{aligned}
\Psi_0 = & \\
& - \frac{2i\,a^5 r^7 \Delta \cos(\theta)\sin(\theta)^4}{a^{10} r^6 \Delta^3 \cos(\theta)^{10} + 2\,a^8 r^8 \Delta^3 \cos(\theta)^8 + a^6 r^{10} \Delta^3 \cos(\theta)^6 + 3\,a^8 r^4 \Delta^3 \cos(\theta)^8 + 6\,a^6 r^6 \Delta^3 \cos(\theta)^6 + 3\,a^4 r^8 \Delta^3 \cos(\theta)^4 + 3\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 6\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 3\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3} \\
& + \frac{i\,a^5 r^6 \Delta \cos(\theta)\sin(\theta)^4 \frac{\partial}{\partial r}\Delta}{a^{10} r^6 \Delta^3 \cos(\theta)^{10} + 2\,a^8 r^8 \Delta^3 \cos(\theta)^8 + a^6 r^{10} \Delta^3 \cos(\theta)^6 + 3\,a^8 r^4 \Delta^3 \cos(\theta)^8 + 6\,a^6 r^6 \Delta^3 \cos(\theta)^6 + 3\,a^4 r^8 \Delta^3 \cos(\theta)^4 + 3\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 6\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 3\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3} \\
& + \frac{2\,a^4 r^6 \Delta^3 \sin(\theta)^4}{a^{10} r^6 \Delta^3 \cos(\theta)^{10} + 2\,a^8 r^8 \Delta^3 \cos(\theta)^8 + a^6 r^{10} \Delta^3 \cos(\theta)^6 + 3\,a^8 r^4 \Delta^3 \cos(\theta)^8 + 6\,a^6 r^6 \Delta^3 \cos(\theta)^6 + 3\,a^4 r^8 \Delta^3 \cos(\theta)^4 + 3\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 6\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 3\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3} \\
& - \frac{2i\,a^3 r^9 \Delta \cos(\theta)\sin(\theta)^2}{a^{10} r^6 \Delta^3 \cos(\theta)^{10} + 2\,a^8 r^8 \Delta^3 \cos(\theta)^8 + a^6 r^{10} \Delta^3 \cos(\theta)^6 + 3\,a^8 r^4 \Delta^3 \cos(\theta)^8 + 6\,a^6 r^6 \Delta^3 \cos(\theta)^6 + 3\,a^4 r^8 \Delta^3 \cos(\theta)^4 + 3\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 6\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 3\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3} \\
& - \frac{2\,a^4 r^8 \sin(\theta)^4}{a^{10} r^6 \Delta^3 \cos(\theta)^{10} + 2\,a^8 r^8 \Delta^3 \cos(\theta)^8 + a^6 r^{10} \Delta^3 \cos(\theta)^6 + 3\,a^8 r^4 \Delta^3 \cos(\theta)^8 + 6\,a^6 r^6 \Delta^3 \cos(\theta)^6 + 3\,a^4 r^8 \Delta^3 \cos(\theta)^4 + 3\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 6\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 3\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3} \\
& - \dots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_1 = {} & \frac{\sqrt{2}a^4 r^4 \Delta^3 \cos(\theta)\sin(\theta)^3}{4\,(a^8 r^4 \Delta^3 \cos(\theta)^8 + 2\,a^6 r^6 \Delta^3 \cos(\theta)^6 + a^4 r^8 \Delta^3 \cos(\theta)^4 + 2\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 4\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} \\
& - \frac{\sqrt{2}a^4 r^5 \cos(\theta)\sin(\theta)^3 \frac{\partial}{\partial r}\Delta}{2\,(a^8 r^4 \Delta^3 \cos(\theta)^8 + 2\,a^6 r^6 \Delta^3 \cos(\theta)^6 + a^4 r^8 \Delta^3 \cos(\theta)^4 + 2\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 4\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} \\
& + \frac{\sqrt{2}a^4 r^4 \Delta^3 \cos(\theta)\sin(\theta)}{4\,(a^8 r^4 \Delta^3 \cos(\theta)^8 + 2\,a^6 r^6 \Delta^3 \cos(\theta)^6 + a^4 r^8 \Delta^3 \cos(\theta)^4 + 2\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 4\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} \\
& + \frac{\sqrt{2}a^2 r^6 \Delta^3 \cos(\theta)\sin(\theta)}{4\,(a^8 r^4 \Delta^3 \cos(\theta)^8 + 2\,a^6 r^6 \Delta^3 \cos(\theta)^6 + a^4 r^8 \Delta^3 \cos(\theta)^4 + 2\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 4\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} \\
& + \frac{\sqrt{2}a^4 r^4 \Delta \cos(\theta)\sin(\theta)^3}{4\,(a^8 r^4 \Delta^3 \cos(\theta)^8 + 2\,a^6 r^6 \Delta^3 \cos(\theta)^6 + a^4 r^8 \Delta^3 \cos(\theta)^4 + 2\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 4\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} \\
& - \frac{i\,\sqrt{2}a^3 r^3 \Delta^3 \sin(\theta)^3}{4\,(a^8 r^4 \Delta^3 \cos(\theta)^8 + 2\,a^6 r^6 \Delta^3 \cos(\theta)^6 + a^4 r^8 \Delta^3 \cos(\theta)^4 + 2\,a^6 r^2 \Delta^3 \cos(\theta)^6 + 4\,a^4 r^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} + \dots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_2 = {} & -\frac{a^4 r^2 \Delta^3 \sin(\theta)^4}{6\,(a^6 r^2 \Delta^3 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^3 \cos(\theta)^4 + a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} + \frac{a^4 r^2 \sin(\theta)^4 \frac{\partial}{\partial r}\Delta^2}{6\,(a^6 r^2 \Delta^3 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^3 \cos(\theta)^4 + a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} \\
& - \frac{a^4 r^2 \Delta \sin(\theta)^4 \frac{\partial^2}{(\partial r)^2}\Delta}{12\,(a^6 r^2 \Delta^3 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^3 \cos(\theta)^4 + a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} + \frac{a^4 r^2 \Delta^3 \sin(\theta)^2}{3\,(a^6 r^2 \Delta^3 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^3 \cos(\theta)^4 + a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} \\
& + \frac{a^2 r^4 \Delta^3 \sin(\theta)^2}{3\,(a^6 r^2 \Delta^3 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^3 \cos(\theta)^4 + a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} - \frac{a^2 r^2 \Delta^4 \sin(\theta)^2}{3\,(a^6 r^2 \Delta^3 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^3 \cos(\theta)^4 + a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} \\
& - \frac{a^4 r^2 \sin(\theta)^2 \frac{\partial}{\partial r}\Delta^2}{3\,(a^6 r^2 \Delta^3 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^3 \cos(\theta)^4 + a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} - \frac{a^2 r^4 \sin(\theta)^2 \frac{\partial}{\partial r}\Delta^2}{3\,(a^6 r^2 \Delta^3 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^3 \cos(\theta)^4 + a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} \\
& + \frac{a^4 r^2 \Delta \sin(\theta)^2 \frac{\partial^2}{(\partial r)^2}\Delta}{6\,(a^6 r^2 \Delta^3 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^3 \cos(\theta)^4 + a^2 r^6 \Delta^3 \cos(\theta)^2 + a^4 \Delta^3 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^3 \cos(\theta)^2 + r^4 \Delta^3)} + \dots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_3 = {} & \frac{\sqrt{2}a^4 r^2 \Delta^3 \cos(\theta)^5}{8\,(a^6 r^2 \Delta^2 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^2 \cos(\theta)^4 + a^2 r^6 \Delta^2 \cos(\theta)^2 + a^4 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^2 \cos(\theta)^2 + r^4 \Delta^2)\sin(\theta)} \\
& - \frac{\sqrt{2}a^4 r^3 \cos(\theta)^5 \frac{\partial}{\partial r}\Delta}{4\,(a^6 r^2 \Delta^2 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^2 \cos(\theta)^4 + a^2 r^6 \Delta^2 \cos(\theta)^2 + a^4 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^2 \cos(\theta)^2 + r^4 \Delta^2)\sin(\theta)} \\
& - \frac{3\,\sqrt{2}a^4 r^2 \Delta^3 \cos(\theta)^3}{8\,(a^6 r^2 \Delta^2 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^2 \cos(\theta)^4 + a^2 r^6 \Delta^2 \cos(\theta)^2 + a^4 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^2 \cos(\theta)^2 + r^4 \Delta^2)\sin(\theta)} \\
& - \frac{\sqrt{2}a^2 r^4 \Delta^3 \cos(\theta)^3}{8\,(a^6 r^2 \Delta^2 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^2 \cos(\theta)^4 + a^2 r^6 \Delta^2 \cos(\theta)^2 + a^4 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^2 \cos(\theta)^2 + r^4 \Delta^2)\sin(\theta)} \\
& + \frac{\sqrt{2}a^4 r^2 \Delta \cos(\theta)^5}{8\,(a^6 r^2 \Delta^2 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^2 \cos(\theta)^4 + a^2 r^6 \Delta^2 \cos(\theta)^2 + a^4 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^2 \cos(\theta)^2 + r^4 \Delta^2)\sin(\theta)} \\
& + \frac{i\,\sqrt{2}a^3 r \Delta^3 \cos(\theta)^4}{8\,(a^6 r^2 \Delta^2 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^2 \cos(\theta)^4 + a^2 r^6 \Delta^2 \cos(\theta)^2 + a^4 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^2 \cos(\theta)^2 + r^4 \Delta^2)\sin(\theta)} \\
& + \frac{\sqrt{2}a^4 r^3 \cos(\theta)^3 \frac{\partial}{\partial r}\Delta}{4\,(a^6 r^2 \Delta^2 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^2 \cos(\theta)^4 + a^2 r^6 \Delta^2 \cos(\theta)^2 + a^4 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^2 \cos(\theta)^2 + r^4 \Delta^2)\sin(\theta)} \\
& - \frac{\sqrt{2}a^2 r^5 \cos(\theta)^3 \frac{\partial}{\partial r}\Delta}{4\,(a^6 r^2 \Delta^2 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^2 \cos(\theta)^4 + a^2 r^6 \Delta^2 \cos(\theta)^2 + a^4 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^2 \cos(\theta)^2 + r^4 \Delta^2)\sin(\theta)} \\
& - \frac{\sqrt{2}a^2 r^4 \Delta^3 \cos(\theta)}{8\,(a^6 r^2 \Delta^2 \cos(\theta)^6 + 2\,a^4 r^4 \Delta^2 \cos(\theta)^4 + a^2 r^6 \Delta^2 \cos(\theta)^2 + a^4 \Delta^2 \cos(\theta)^4 + 2\,a^2 r^2 \Delta^2 \cos(\theta)^2 + r^4 \Delta^2)\sin(\theta)} - \dots
\end{aligned}
$$

$$\Psi_4 = -\frac{i\,a^5 r^3 \Delta \cos(\theta)^5}{2\,(a^6 r^2 \Delta \cos(\theta)^6 + 2\,a^4 r^4 \Delta \cos(\theta)^4 + a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)} + \frac{i\,a^5 r^2 \Delta \cos(\theta)^5 \frac{\partial}{\partial r} \Delta}{4\,(a^6 r^2 \Delta \cos(\theta)^6 + 2\,a^4 r^4 \Delta \cos(\theta)^4 + a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)}$$
$$+ \frac{a^4 r^2 \Delta^3 \cos(\theta)^4}{2\,(a^6 r^2 \Delta \cos(\theta)^6 + 2\,a^4 r^4 \Delta \cos(\theta)^4 + a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)} + \frac{i\,a^5 r^3 \Delta \cos(\theta)^3}{a^6 r^2 \Delta \cos(\theta)^6 + 2\,a^4 r^4 \Delta \cos(\theta)^4 + a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta}$$
$$+ \frac{i\,a^3 r^5 \Delta \cos(\theta)^3}{2\,(a^6 r^2 \Delta \cos(\theta)^6 + 2\,a^4 r^4 \Delta \cos(\theta)^4 + a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)} - \frac{a^4 r^4 \cos(\theta)^4}{2\,(a^6 r^2 \Delta \cos(\theta)^6 + 2\,a^4 r^4 \Delta \cos(\theta)^4 + a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)}$$
$$+ \frac{i\,a^3 r^4 \Delta \cos(\theta)^3 \frac{\partial}{\partial r} \Delta}{4\,(a^6 r^2 \Delta \cos(\theta)^6 + 2\,a^4 r^4 \Delta \cos(\theta)^4 + a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)} - \frac{i\,a^3 r^3 \Delta^2 \cos(\theta)^3}{a^6 r^2 \Delta \cos(\theta)^6 + 2\,a^4 r^4 \Delta \cos(\theta)^4 + a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta}$$
$$+ \frac{a^4 r^2 \Delta \cos(\theta)^4}{a^6 r^2 \Delta \cos(\theta)^6 + 2\,a^4 r^4 \Delta \cos(\theta)^4 + a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta} - \frac{a^4 r \Delta \cos(\theta)^4 \frac{\partial}{\partial r} \Delta}{2\,(a^6 r^2 \Delta \cos(\theta)^6 + 2\,a^4 r^4 \Delta \cos(\theta)^4 + a^2 r^6 \Delta \cos(\theta)^2 + a^4 \Delta \cos(\theta)^4 + 2\,a^2 r^2 \Delta \cos(\theta)^2 + r^4 \Delta)} + \dots$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"